

Healthcare Cancer Survival Analysis

An End-to-End Healthcare Analytics Project Using SQL, Power BI, and R

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Introduction

Cancer is a major public health challenge and continues to affect millions of people worldwide. Understanding the factors associated with patient survival is important for healthcare providers because it can support better clinical decisions and improve patient outcomes. As healthcare organizations collect more patient data, analytical tools have become increasingly valuable for identifying patterns and supporting evidence-based decision-making.

In this project, SQL, Power BI, and R were used to analyze a dataset of cancer patients. The analysis followed a complete analytics process, beginning with data validation and cleaning, followed by analysis, dashboard development, regression modeling, and model validation. Rather than focusing only on statistical results, the project demonstrates how multiple analytical tools can be integrated to answer real healthcare questions.

The goal of this project is to show how healthcare data can be organized, analyzed, and presented in a way that supports both clinical and operational decision-making while demonstrating practical skills in business intelligence and healthcare analytics.

2. Research Objective

Primary Objective

To identify the demographic and clinical factors associated with cancer patient survival using statistical modeling and healthcare analytics techniques.

Secondary Objectives

- Validate and clean healthcare data using SQL.
- Develop an interactive Power BI dashboard for exploratory analysis.
- Evaluate relationships between patient characteristics and survival outcomes using multiple linear regression.
- Assess regression assumptions through diagnostic testing.

- Translate analytical findings into practical healthcare recommendations.

Dataset Description

Dataset

Source: India Cancer Patient Dataset (2022–2025)-Kaggle

Observations: 100,000 patients

Variables

Variable	Description
Patient_ID	Unique patient identifier
Age	Age in years
Gender	Male/Female
State	Patient state
City	Patient city
Hospital_Name	Treating hospital
Cancer_Type	Type of cancer
Stage	Cancer stage (I–IV)
Treatment_Type	Treatment received
Diagnosis_Date	Date of diagnosis
Survival_Months	Survival duration
Status	Patient outcome/status

Methodology

This section explains **how** we conducted the analysis.

Step 1 – SQL

- Imported the dataset into SQL Server.
- Verified data integrity.
- Checked for missing values.
- Identified duplicate records.
- Produced descriptive statistics.
- Answered exploratory business questions.

Step 2 – Power BI

- Connected Power BI to SQL Server.
- Built interactive dashboards.
- Developed KPIs.
- Created visualizations for demographics, cancer types, treatment distribution, stage distribution, and patient status.

Step 3 – R

- Imported the cleaned SQL dataset.
- Conducted exploratory analysis.
- Developed three regression models.
- Calculated confidence intervals.
- Evaluated multicollinearity using VIF.
- Performed regression diagnostic analysis.

SQL Analysis

Data Validation and Cleaning

Prior to conducting statistical analysis, the dataset was imported into Microsoft SQL Server and subjected to a comprehensive data validation process. Data quality checks were performed to ensure the reliability and integrity of the analysis.

The validation process included:

- Verification of missing values across all variables.
- Identification of duplicate patient records using the unique Patient_ID.
- Validation of numerical variables, including age and survival months.
- Examination of categorical variables such as gender, cancer stage, treatment type, and cancer type.

The dataset contained **100,000 patient records** with no duplicate observations and no missing values in the variables used for analysis. This provided a high-quality dataset suitable for analysis and statistical modeling.

Exploratory SQL Analysis

SQL queries were developed to summarize key characteristics of the patient population and identify initial trends before statistical modeling.

The exploratory analysis included:

- Distribution of patients by gender
- Distribution of cancer types
- Distribution of treatment types
- Distribution of cancer stages
- Patient status frequencies
- Average age
- Average survival months
- Geographic distribution by state
- Cross-tabulation of cancer type and gender

These exploratory summaries helped identify patterns that were later investigated using regression analysis.

Key SQL Findings

Based on our SQL analysis:

The dataset consisted of approximately **67% female** and **33% male** patients. The average patient age was approximately **54 years**, while the average survival time was approximately **20 months**.

Stage IV represented a substantial proportion of patients, suggesting that many diagnoses occurred after significant disease progression.

Exploratory analysis also revealed variation in survival across cancer stages and cancer types, motivating the development of regression models to quantify these relationships while controlling for multiple patient characteristics.

Power BI Dashboard

Dashboard Overview

An interactive Power BI dashboard was developed to enable rapid exploration of demographic and clinical characteristics within the dataset. The dashboard was designed to support healthcare decision-makers by presenting key performance indicators alongside interactive visualizations.

The dashboard included:

- Total Patients
- Male Patients
- Female Patients
- Average Age
- Average Survival Months

Interactive visuals included:

- Cancer Type by Gender
- Patient Distribution by State
- Treatment Distribution
- Cancer Stage Distribution
- Patient Status Distribution
- State Slicer

Purpose of the Dashboard

The dashboard provides decision-makers with an intuitive overview of patient demographics, disease characteristics, and treatment patterns. Interactive filtering allows users to explore regional differences and identify trends that may require further investigation.

Dashboard Insights

Our dashboard highlighted several important observations.

Finding 1

The patient population contained more female than male patients.

Finding 2

Stage III and Stage IV cancers represented a significant proportion of diagnoses.

Finding 3

Cancer types varied substantially across the patient population, suggesting the need for disease-specific analyses.

Finding 4

Treatment distributions showed that patients received a variety of treatment approaches, reflecting differences in clinical presentation and disease severity.

Finding 5

Interactive geographic analysis enabled comparison of patient populations across states.

Regression Analysis

Objective

Multiple linear regression was performed to evaluate the association between demographic and clinical characteristics and patient survival.

Three models were estimated.

Model 1

Predictors

Age

Results

Age demonstrated only a very small negative association with survival months.

The model explained virtually none of the variation in patient survival.

This suggested that age alone is not a meaningful predictor of survival outcomes within this dataset.

Model 2

Predictors

- Age
- Cancer Stage

Results

Adding cancer stage substantially improved model performance.

Compared with Stage I patients,

- Stage II patients experienced significantly lower expected survival.
- Stage III patients experienced even larger reductions.
- Stage IV patients demonstrated the greatest reduction in survival.

Cancer stage emerged as the dominant predictor.

Model 3

Predictors

- Age
- Gender
- Cancer Stage
- Cancer Type
- Treatment Type

Results

The full regression model produced the strongest explanatory performance among the three models.

The analysis demonstrated:

- Cancer stage remained the strongest predictor.

- Several cancer types showed statistically significant differences in survival.
- Gender had minimal independent influence.
- Treatment type was not statistically significant after adjustment for the other variables.

Model Validation

Correlation Analysis

Prior to regression modeling, a Pearson correlation analysis was conducted to examine the relationship between the numerical variables included in the analysis.

The correlation between **Age** and **Survival Months** was **-0.02**, indicating a negligible negative linear relationship. This finding suggested that patient age alone explains very little variation in survival outcomes.

Confidence Intervals

Ninety-five percent confidence intervals were calculated for all regression coefficients to assess the precision of the estimated effects.

The confidence intervals for the cancer stage variables did not include zero, confirming that Stage II, Stage III, and Stage IV were statistically significant predictors of survival. In contrast, several treatment categories produced confidence intervals that included zero, indicating that their estimated effects were not statistically significant after controlling for the remaining predictors.

Multicollinearity Assessment

Multicollinearity was evaluated using the Variance Inflation Factor (VIF). All adjusted GVIF values were close to **1.0**, indicating that the explanatory variables were not highly correlated with one another.

The absence of problematic multicollinearity suggests that the regression coefficients are stable and can be interpreted with confidence.

Residual Diagnostics

Regression diagnostic plots were examined to evaluate the assumptions of linear regression.

The residuals were randomly distributed around zero with no strong systematic patterns. The Scale-Location plot suggested relatively constant residual variance, while the Residuals versus Leverage plot indicated no highly influential observations. Although the

Normal Q-Q plot showed mild departures from normality in the tails, the large sample size (100,000 observations) supports the robustness of the regression estimates for statistical inference.

Key Findings

The analysis produced several important findings.

Finding 1

Cancer stage was the strongest predictor of patient survival.

Patients diagnosed at later stages experienced substantially lower survival months compared with Stage I patients.

Finding 2

Age demonstrated only a weak relationship with survival.

Although age appeared statistically significant in the simple regression model, its influence became much smaller after controlling for disease characteristics.

Finding 3

Cancer type contributed meaningful explanatory power.

Several cancer types exhibited significantly different survival outcomes compared with the reference category, highlighting the importance of disease-specific analysis.

Finding 4

Gender was not a strong independent predictor of survival after accounting for cancer stage and cancer type.

Finding 5

Treatment type was not statistically significant in the final model after adjustment for the remaining explanatory variables.

This finding should not be interpreted as evidence that treatment is ineffective. Rather, it suggests that treatment type alone did not explain additional variation in survival beyond the other variables included in the model.

Finding 6

The regression diagnostics indicated that the statistical assumptions were reasonably satisfied, supporting the reliability of the analytical results.

Healthcare Recommendations

Based on the findings, several recommendations can be made.

Recommendation 1: Strengthen Early Detection Programs

Cancer stage emerged as the strongest predictor of survival. Increasing access to screening and early diagnosis may improve patient outcomes by identifying disease before it progresses to advanced stages.

Recommendation 2: Prioritize High-Risk Patient Groups

Healthcare providers should identify patients with advanced-stage disease for closer monitoring, coordinated care, and timely intervention.

Recommendation 3: Use Interactive Dashboards for Decision Support

Healthcare organizations can use business intelligence dashboards to monitor patient demographics, treatment patterns, and disease distributions in real time, supporting operational planning and resource allocation.

Recommendation 4: Expand Data Collection

Future analyses should include additional clinical variables such as:

- Tumor grade
- Comorbidities
- Smoking status
- Treatment duration
- Genetic markers
- Follow-up outcomes

These variables may improve the predictive performance of future statistical models.

Study Limitations

Limitation 1

The analysis is based on observational data and therefore identifies statistical associations rather than causal relationships.

Limitation 2

Several clinically relevant variables were unavailable, limiting the amount of variation explained by the regression models.

Limitation 3

The final regression model explained only a modest proportion of the variability in survival outcomes, suggesting that additional patient and clinical characteristics not included in the dataset influence survival.

Limitation 4

The findings are specific to the available dataset and should be interpreted cautiously when applying them to different healthcare systems or patient populations.

Conclusion

This project used SQL, Power BI, and R to analyze factors associated with cancer patient survival. SQL was used to clean, validate, and prepare the data for analysis. Power BI was used to create interactive dashboards that summarized patient demographics, cancer characteristics, and treatment patterns, while R was used to build and evaluate multiple regression models.

The analysis found that cancer stage was the strongest predictor of patient survival. Although age showed a relationship with survival in the initial analysis, its effect became much smaller after accounting for cancer stage and other patient characteristics. Gender also had little independent influence on survival. Regression diagnostics indicated that the model met the necessary assumptions, providing confidence in the overall results.

This project also strengthened practical skills in data cleaning, visualization, statistical analysis, and communicating findings. It demonstrates how SQL, Power BI, and R can be used together to analyze healthcare data and produce insights that support data-informed decision-making.